

SPHEROID METRICS

Anonymous Author(s)

Abstract

Pancreatic cancer remains highly lethal and difficult to treat, motivating the use of three-dimensional spheroid models that accurately capture tumor growth and drug response compared to conventional monolayer cultures. Variability in spheroid shape, imaging conditions, and debris further limits the reliability and reproducibility of manual and semi-automated approaches, which are time-consuming and subjective. Despite these challenges, quantitative morphological metrics derived from these spheroids provide critical insight into treatment effects.

This work presents an automated image-analysis pipeline for spheroid segmentation and morphological quantification using bright-field microscopy spheroid images. The approach employs deep learning-based segmentation models, including YOLOv8 segmentation and U-Net, to robustly delineate spheroid boundaries under challenging imaging conditions. Following segmentation, the pipeline extracts key morphological and intensity-based metrics such as relative area, circularity, and gray-level co-occurrence matrix (GLCM) features to enable comparatively fast and scalable assessment of tumor growth and drug efficacy in 3D cancer models. **Keywords:** Three Dimensional(3D) Spheroids; In-vivo; In-vitro; Controlled; Treated; Segmentation Model; Morphological Metrics

Code: https://github.com/thws-mai/PM25_Efficacy

1 Introduction

Pancreatic cancer remains one of the deadliest malignancies, characterized by late diagnosis, rapid progression, and limited therapeutic response. Preclinical evaluation of drug efficacy increasingly relies on three-dimensional (3D) tumor spheroid models grown in vitro (outside a living organism under controlled laboratory conditions), which more accurately reflect aspects of the in vivo tumor microenvironment (within a living organism) than traditional two-dimensional cultures. Spheroids preserve critical biological characteristics, including cell-cell interactions, nutrient gradients, hypoxia, and drug penetration profiles, making them well suited for studying tumor growth dynamics and treatment response.

Quantitative analysis of spheroid morphology from bright-field microscopy images plays a central role in assessing drug-induced effects such as growth inhibition, structural degradation, and changes in cellular viability. Reliable measurement of these effects requires accurate segmentation of spheroid boundaries to separate tumor regions from background and imaging artifacts. However, manual annotation and measurement remain labor-intensive, subjective, and difficult to reproduce, particularly in longitudinal and high-throughput studies. Variability in imaging conditions, irregular spheroid morphologies, debris, and low contrast between spheroids and the background further complicate consistent segmentation and metric extraction, limiting scalability and reproducibility.

Recent advances in deep learning-based image segmentation provide effective tools for automating spheroid boundary detection and enabling systematic morphological analysis. Nevertheless, existing approaches often exhibit important limitations. Many recent

pipelines rely on fluorescence imaging to infer viability or cell composition, which increases experimental complexity and requires labeling and staining. Other methods focus primarily on detection or segmentation and leave quantitative analysis to the user, while some end-to-end convolutional models learn abstract features that are difficult to interpret biologically. These limitations restrict the adoption of automated spheroid analysis in routine bright-field assays.

In this work, we present an end-to-end automated pipeline for spheroid segmentation and quantitative analysis in bright-field microscopy images using YOLOv8 segmentation and U-Net architectures. The pipeline integrates robust data pre-processing and augmentation, enables accurate spheroid boundary delineation under challenging imaging conditions, and supports automated extraction of biologically relevant morphological, intensity, and texture-based metrics to quantify spheroid growth changes and drug-induced effects. The proposed framework serves as a baseline, robust, and interpretable analysis tool for bright-field spheroid assays and functions as a first-line screening approach prior to more complex or modality-specific analyses, supporting scalable and reproducible evaluation in preclinical cancer studies.

2 Related work

Early approaches to spheroid image analysis rely on manual annotation or classical image processing techniques. Manual measurement provides direct control but remains subjective and impractical for large datasets or longitudinal studies. Classical methods based on thresholding or edge detection offer partial automation but struggle under variable imaging conditions, irregular spheroid morphologies, and low contrast commonly observed in bright-field microscopy images.

Recent advances in deep learning significantly improve spheroid detection and segmentation performance. Convolutional neural networks, particularly U-Net-based architectures, demonstrate strong capability in biomedical image segmentation by capturing multiscale spatial features. Object detection and segmentation frameworks such as YOLO further enable efficient localization and boundary delineation of spheroids in complex scenes. These learning-based methods improve robustness compared to traditional approaches and support automated processing across diverse datasets.

However, most recent deep learning-based pipelines address only limited aspects of spheroid analysis. Many approaches operate on fluorescence microscopy images to predict cell viability or population fractions, which increases experimental complexity and requires labeling. Other methods focus primarily on spheroid detection or segmentation and leave quantitative morphological analysis entirely to the user. Several end-to-end convolutional models produce global predictions or learned feature representations that are difficult to interpret biologically.

These limitations highlight the need for automated spheroid analysis frameworks that operate directly on bright-field images, integrate robust segmentation with interpretable metric extraction,

and support scalable and reproducible analysis. The proposed work builds upon recent deep learning advances while addressing practical gaps in usability, interpretability, and quantitative assessment for preclinical cancer studies.

3 Dataset and Preprocessing

3.1 Dataset Description

We use bright-field microscopy images of pancreatic cancer spheroids acquired under controlled and drug-treated conditions. The primary dataset contains 179 fully annotated images with YOLO-format labels, which we use to train and validate the segmentation models. We verify all annotations to confirm that each image contains valid spheroid labels.

We use a second dataset consisting of controlled and treated spheroid images without annotations. We reserve this dataset exclusively for inference, metric comparison, and biological interpretation, and we do not use it during model training or validation.

3.2 Annotation Conversion and Organization

We convert YOLO-format polygon annotations into binary segmentation masks that are compatible with U-Net training. We organize images and corresponding masks into structured directories to ensure reproducibility and support collaborative workflows. We match each image-mask pair using consistent filename conventions to prevent misalignment during data loading and training.

3.3 Preprocessing Pipeline and Data Augmentation

The limited availability of annotated training data constrains robust deep learning-based segmentation. To mitigate this limitation and improve model generalization, we apply a controlled data augmentation strategy during training. Specifically, we perform horizontal and vertical flipping, as well as rotations by 90° and 270°. These transformations preserve spheroid morphology while increasing the effective diversity of the training dataset.

The preprocessing pipeline includes the following steps:

- (1) Matching images from the image directory with corresponding segmentation masks from the label directory.
- (2) The dataset is partitioned into training, validation, and test sets directories with the ratio of 70/15/15 respectively using filename suffix-based grouping to ensure that augmented versions of the sample remain within the respective subset.
- (3) Resizing images and masks to fixed spatial dimensions (512 × 512 for U-Net or 768 × 768 for Yolov8n) using uniform scaling
- (4) Pixel intensity normalization using mean and standard deviation calculated from the dataset images with the custom code.
- (5) Batch loading for efficient model training.

3.4 Preprocessing of Unannotated Experimental Data

A separate preprocessing pipeline is applied to the unannotated experimental dataset, which includes spheroid images acquired under varying magnification, framing, and illumination conditions. These

differences introduce additional challenges for direct inference and metric extraction.

To address these variations, the following steps are applied exclusively to the unannotated dataset:

- (1) Automated spheroid localization using coarse model inference combined with intensity-based heuristics.
- (2) Centered zoom-cropping around detected spheroid regions to remove excess background and peripheral debris, followed by rescaling to match the spatial resolution used during training.

These preprocessing operations are strictly isolated from the training and validation pipelines to preserve the integrity of model evaluation.

4 Methodology

4.1 Design Objectives and Challenges

The objective of this work is to achieve accurate spheroid boundary segmentation that enables reliable extraction and comparison of morphological and intensity-based metrics for assessing spheroid growth changes and drug-induced effects. This task is constrained by a limited amount of annotated training data, the presence of debris and low-contrast backgrounds, irregular and evolving spheroid morphologies, and variability in image acquisition conditions between control and treated samples. These challenges require segmentation models that are robust to noise, preserve boundary fidelity, and generalize well under limited supervision.

To address these constraints, we investigate two complementary deep learning based segmentation approaches such as YOLOv8 segmentation and a U-Net architecture and evaluate their suitability for downstream biological analysis. We perform model selection based on segmentation quality, as discussed in Section 6.

4.2 YOLOv8-Based Instance Segmentation

We train a YOLOv8n-seg model to perform automated spheroid detection and instance-level segmentation on bright-field microscopy images. YOLOv8 is selected for its unified detection-segmentation formulation, high inference efficiency, and robustness in cluttered scenes under limited annotated data. These properties are particularly important for spheroid images containing debris, merged objects, and heterogeneous backgrounds.

The model is trained for 1000 epochs using the Ultralytics YOLOv8 training pipeline with an input resolution of 768 × 768. Batch size is selected based on available hardware resources. Conservative, boundary-preserving augmentations are applied during training, including mild Gaussian blur ($p = 0.20$), while aggressive intensity normalization and brightness or contrast jitter are avoided to preserve boundary consistency in low-contrast regions.

$$L = L_{\text{box}} + L_{\text{cls}} + L_{\text{mask}} + L_{\text{dfl}}, \quad (1)$$

YOLOv8 optimizes a composite loss function defined as in the equation 1, which jointly enforces accurate object localization, classification, and mask prediction. This formulation encourages coherent instance masks and stable boundary estimation, which are essential for downstream morphological measurements such as area and perimeter. YOLOv8 provides an efficient first-stage segmentation approach and serves as a strong candidate for robust

spheroid localization and mask generation under challenging imaging conditions.

4.3 U-Net–Based Pixel-Wise Segmentation

To achieve higher boundary precision and shape preservation, we employ a U-Net–based fully convolutional network for pixel-wise spheroid segmentation. The architecture follows a standard encoder–decoder design with symmetric skip connections, enabling accurate spatial localization while retaining contextual information. This design is critical for spheroid segmentation, where small boundary errors directly affect downstream metrics such as perimeter, circularity, and area.

Given the limited availability of annotated training data, a ResNet34 encoder with pretrained weights is employed to improve feature generalization and mitigate overfitting. The U-Net model is trained for 30 epochs using images resized to 512×512 pixels, a resolution that balances boundary detail preservation with computational efficiency. We perform optimization using the Adam optimizer with a learning rate of 1×10^{-4} , providing stable convergence during training.

Accurate segmentation in bright-field images requires explicit supervision of boundary alignment and size preservation. Therefore, we employ a multi-term loss function combining pixel-wise classification loss, an IoU-based overlap loss, a boundary IoU loss to penalize edge misalignment, and an area-based loss to discourage over- and under-segmentation. This formulation directly addresses challenges mentioned in 4.1 ensuring that segmentation outputs remain biologically meaningful and suitable for metric extraction.

$$L = \alpha_{\text{BCE}} L_{\text{BCE}} + \alpha_{\text{IoU}} L_{\text{IoU}} + \alpha_{\text{Boundary}} L_{\text{Boundary IoU}} + \alpha_{\text{Area}} L_{\text{Area}} \quad (2)$$

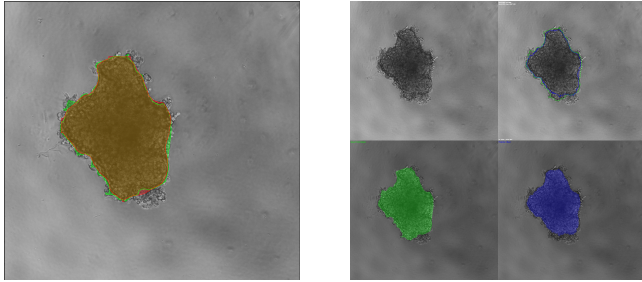


Figure 1: Comparison of YOLO (left) and UNET (right) results

YOLOv8 emphasizes robust instance-level segmentation under cluttered and heterogeneous imaging conditions, while U-Net prioritizes fine-grained boundary accuracy and shape preservation. Both models generate segmentation masks that serve as the basis for automated extraction of morphological descriptors and pixel intensity statistics. Comparative evaluation of segmentation performance and downstream metric consistency is used to select the most suitable model for morphological and intensity-based metric measurement.

4.4 Postprocessing

To improve the reliability of downstream metric extraction, predicted segmentation masks are post-processed to suppress isolated regions and enforce structural consistency. In a small subset of treated samples, low contrast and dark background regions occasionally produce fragmented or weakly connected predictions, particularly near indistinct spheroid boundaries. Morphological opening and closing operations reduce such artifacts, while connected-component filtering removes small, non-spheroid regions, ensuring that retained masks correspond to contiguous and biologically plausible spheroid structures.

4.5 Automated Analysis Pipeline

The complete automated analysis pipeline integrates segmentation and quantitative analysis to support spheroid growth and drug-response studies. The pipeline consists of the following stages:

- (1) Data ingestion and preprocessing.
- (2) Model inference to generate spheroid segmentation masks.
- (3) Post-processing to remove artifacts and isolate valid spheroid regions.
- (4) Feature extraction from segmented spheroids.
- (5) Statistical comparison between control and drug-treated samples.

Steps (1) through (3) are addressed as part of the previous section 3 and 4 (this section), including data preparation, model inference, and mask post-processing. Steps (4) and (5), which focus on morphological and intensity-based metric extraction and comparative analysis, are described in detail in the subsequent section 5.

5 Feature Extraction and Metrics

Quantitative assessment of spheroid growth and drug response requires features that are both biologically interpretable and robust to imaging variability. We extract complementary sets of morphological and texture-based metrics from the segmented spheroid regions.

5.1 Morphological Metrics

Morphological metrics are extracted from the segmented spheroid masks to quantify changes in spheroid size, shape, and structural integrity following drug treatment. These features provide direct and interpretable indicators of growth suppression and cytotoxic effects and are widely used in spheroid-based assays.

Area measures the total number of pixels within the segmented spheroid region and serves as the primary indicator of spheroid size. Differences in area between control and treated samples reflect growth inhibition, cell loss, or spheroid disintegration. It is relatively robust to minor boundary irregularities and is computed as $A = \sum_{x,y} M(x,y)$, where $M(x,y) \in \{0,1\}$ denotes the binary segmentation mask. In addition to size, perimeter characterizes boundary length and captures edge complexity and surface irregularity; an increased perimeter without a proportional increase in area indicates boundary degradation or reduced compactness following treatment. Complementarily, circularity quantifies deviation from an ideal circular shape and is defined as, $\text{Circularity} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$. Lower circularity values correspond to irregular, elongated, or

fragmented spheroids and indicate reduced structural stability. Together, these metrics capture size, boundary complexity, and shape compactness, enabling objective comparison between control and treated conditions.

5.2 Texture Metrics Using Gray-Level Co-occurrence Matrix (GLCM)

While morphological metrics describe global shape changes, they do not capture internal intensity variations that reflect spheroid heterogeneity and treatment-induced disruption. To address this, we extract texture features using the Gray-Level Co-occurrence Matrix (GLCM), which characterizes spatial intensity relationships within the spheroid region and is robust to global brightness variations.

Given a quantized grayscale image with N_g gray levels, the GLCM $P(i, j | d, \theta)$ represents the normalized co-occurrence frequency of intensity levels i and j at pixel distance d and orientation θ . Each image is converted to grayscale, quantized to $N_g = 32$ levels, and analyzed only within the predicted spheroid region. To reduce boundary-related artifacts, the spheroid mask is eroded by three pixels, and masked-out pixels are excluded from computation.

Symmetric, normalized GLCMs are computed for distances $d \in \{1, 2, 4\}$ and orientations $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$. Texture metrics are reported as the mean across all (d, θ) combinations. We extract Energy ($\sum_{i,j} P(i, j)^2$), Entropy ($-\sum_{i,j} P(i, j) \log_2(P(i, j) + \epsilon)$), and Contrast ($\sum_{i,j} (i - j)^2 P(i, j)$), which collectively capture uniformity, randomness, and local intensity variation within the spheroid.

Because GLCM-based features depend on relative intensity co-occurrence patterns rather than absolute brightness values, they enable consistent comparison between control and treated spheroids under varying illumination and acquisition conditions.

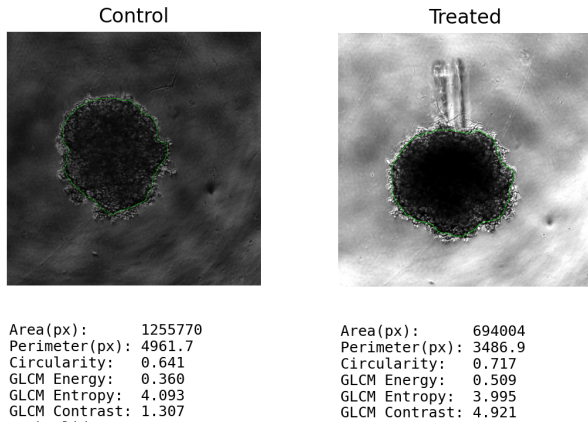


Figure 2: Comparison of Control and Treated metrics results

6 Experimental Results and Evaluation

Segmentation performance is evaluated to determine suitability for stable and interpretable metric extraction rather than to maximize segmentation accuracy alone. Evaluation focuses on region overlap or IoU = $\frac{|P \cap G|}{|P \cup G|}$ (where P = predicted mask and G = ground truth mask), boundary fidelity and area preservation, which directly affect morphological and intensity-based measurements.

YOLOv8 achieves a mean IoU of 92% and an average ground truth-to-prediction area ratio of 99%, indicating accurate spheroid localization and strong size preservation across control and treated samples, while remaining robust in cluttered and low-contrast conditions. The U-Net model attains a mean IoU of 0.90, demonstrating strong region overlap and smoother boundaries in high-contrast cases; however, boundary ambiguity under low contrast and occasional misalignment affect perimeter-sensitive measurements, reflected in reduced Boundary IoU and increased variability in area agreement. Given the emphasis on reliable region isolation and metric stability, YOLOv8 is selected for subsequent analysis, prioritizing robustness and consistency over marginal boundary smoothness improvements.

For comparative analysis between control and treated spheroids, only circularity is considered from the morphological category, and texture-based metrics derived from the Gray-Level Co-occurrence Matrix as these metrics are unitless and normalized that mitigate variability introduced by time-dependent and environmental factors, such as changes in illumination during image acquisition.

Overall, the evaluation confirms that reliable segmentation directly supports stable extraction of interpretable, unitless features, enabling consistent assessment of spheroid growth changes and drug-induced effects in bright-field assays.

7 Conclusion and Future Work

This work presents an automated pipeline for spheroid segmentation by integrating deep learning-based methods with morphological and GLCM-based feature extraction, the framework enables reproducible and interpretable assessment of spheroid growth/ drug induced changes under limited annotated data and heterogeneous imaging conditions. The selected YOLOv8-based segmentation provides reliable region isolation and consistent metric preservation, forming the basis for objective comparison between control and treated spheroids. While circularity and texture descriptors provide meaningful morphological signatures, they do not capture complex phenotypes such as subtle invasion patterns or multi-layer structures; thus, drug efficacy conclusions remain indirect and morphology-driven rather than mechanistic. The proposed system therefore serves as a baseline, robust, and interpretable first-line screening tool for bright-field spheroid assays.

The current implementation is designed for technically proficient stakeholders capable of operating the raw analysis pipeline. Future work will focus on developing a user-friendly interface to enable direct use by researchers in preclinical drug evaluation studies and on extending the framework to support longitudinal spheroid analysis across multiple time points.

References

- F. Bresciani et al., "A brightfield cell viability model using deep learning," *Cells*, 2019.
- R. Ravi et al., "Deep learning pipeline for morphological and viability analysis," *Critical Reviews in Oncology/Hematology*, 2015.
- A. Friedrich et al., "Spheroid-based tumor models for drug testing," *Nature Protocols*, 2009.
- M. Streller et al., "Image segmentation of treated and untreated tumor spheroids by fully convolutional networks," *arXiv preprint*, 2024.
- A. Akshay et al., "SpheroScan: a user-friendly deep learning tool for spheroid image analysis," *GigaScience*, 2023.
- A. Torkamannia et al., "SYNDEEP: a deep learning approach for the prediction of cancer drugs synergy," *Scientific Reports*, vol. 13, 2023.